34% of carts were abandoned, and the model finds who will buy

Measured results from the full run: 3.6 million clickstream events, 535 thousand orders and 200 thousand customers (synthetic data). Your own 0.02-scale run shows the same pattern at smaller numbers.

94,218
abandoned carts out of 277,754

25.6M
value left in abandoned carts

0.834
model AUC (target 0.75)

0.755
model F1 (target 0.65)

9 of 9
reconciliation checks passed

Shopping funnel

Sessions that reached each stage. Abandoned means a cart with no checkout or purchase.

All sessions	699,980
Added to cart	277,754
Purchased	133,171
Abandoned cart	94,218

What drives a purchase

Top five signals in the Random Forest, from session behaviour and customer history. Precision 0.67, recall 0.86.

Cart value	16.4%
Loyalty tier	15.5%
Desktop device	13.8%
Hour of day	12.0%
Mobile device	9.9%

Customer segments

K-Means with 5 groups, chosen by silhouette score 0.505.

Segment	Customers	Av
Champions	38,561	
Loyal	29,869	
Promising	74,822	
Never bought (two groups)	56,748	



Data quality and reconciliation

3,602,945 raw lines: 36,150 malformed were set aside and 17,702 duplicates removed, leaving 3,549,093 clean rows.

Check	Result	Status
Raw lines equal parsed plus malformed	3,602,945	PASS
Parsed equal clean plus duplicates	3,566,795	PASS
Duplicates removed equal duplicates injected	17,702	PASS
Purchase events matched to orders	133,171	PASS
Order totals equal sum of items	0 mismatches	PASS
Gold customers and lifetime value equal source	200,000 / 191.3M	PASS

Alert speed and what is not measured

Alert delay is a model of the streaming job, not a live measurement. Without a heartbeat, quiet hours stall the alerts.

35 s
modeled p95 delay

370 s
modeled p99 without heartbeat

10–15 s
modeled delay with heartbeat

Not measured, because they need a live system or a real campaign: **18% cart recovery** **+12% lifetime value** **HBase lookup under 100 ms** **live 60 s alert delay**